Practice 14 Solution (C++)

- Two Sum
  - A hash table (or a set) contains integers that have been visited so far
  - Check if the current integer and any of the integers in the hash table add up to k
- Symmetric Pairs
  - A hash table (or a set) contains integer pairs that have been visited so far
  - Check if the current pair is symmetric to any of the visited pairs in the hash table

```cpp
bool twoSum(int k, vector<int>& v) {
  unordered_map<int, bool> visited;
  unordered_map<int, bool>::iterator it;
  for (int i = 0; i < v.size(); ++i) {
    it = visited.find(k - v[i]);
    if (it != visited.end())
      return true;
    visited[v[i]] = true;
  }
  return false;
}

struct pair_hash {
  std::size_t operator () (const std::pair<int, int>& p) const {
    return p.first ^ p.second;
  }
};

typedef unordered_map<pair<int, int>, bool, pair_hash> hash_t;
vector<pair<int, int>> symmetricPairs(vector<pair<int, int>>& arr) {
  hash_t h;
  hash_t::iterator hit;
  vector<pair<int, int>> out;
  vector<pair<int, int>>::iterator it = arr.begin();
  for (; it != arr.end(); ++it) {
    hit = h.find(make_pair(it->second, it->first));
    if (hit != h.end())
      out.push_back(*it);
    else
      h[*it] = true;
  }
  return out;
}
```